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TEAM NAME: HASH_HACKERS

COLLEGE NAME: SRI VENKATESWARA COLLEGE OF ENGINEERING

PROBLEM STATEMENT: DIGITAL EVIDENCE INTEGRITY SYSTEM

PROBLEM STATEMENT

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Digital Evidence Management Challenges

- Modern investigations rely heavily on digital evidence such as CCTV footage, forensic reports, and investigation documents.
- Current evidence management systems are centralized and lack reliable mechanisms to ensure file integrity, transparent chain-of-custody tracking, and automated verification.
- This creates risks of undetected tampering, poor accountability during evidence transfers, and difficulty proving authenticity in legal proceedings.
- Therefore, there is a critical need for a secure system that ensures integrity, traceability, and verifiable management of digital evidence.

1 Evidence Integrity and Tampering Risks

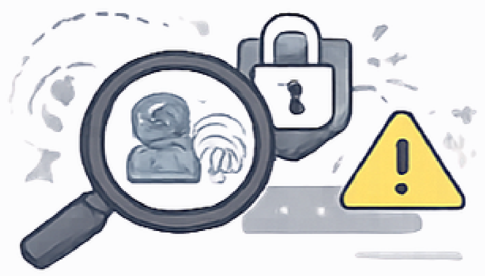
- Digital evidence such as CCTV footage, forensic reports, and investigation documents are often stored in centralized systems.
- Even a minor alteration can compromise the credibility of evidence during legal proceedings.

2 Lack of Transparent Chain-of-Custody Tracking

- Evidence frequently passes through multiple authorities such as police officers, forensic analysts, prosecutors, and court officials.
- Traditional systems lack clear records of who accessed, transferred, or handled the evidence, making accountability and traceability difficult.

3 Absence of Reliable Verification and Monitoring

- Existing evidence management systems do not provide reliable mechanisms to verify file authenticity or detect tampering automatically.
- Without cryptographic verification and real-time monitoring, investigators cannot easily confirm whether evidence has remained unchanged.



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SOLUTION

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A secure platform that protects digital evidence using cryptographic hashing, blockchain technology, intelligent automation, and real-time monitoring to ensure authenticity and transparency throughout the investigation lifecycle.

Cryptographic Evidence Hashing

Each uploaded evidence file generates a SHA-256 cryptographic hash, creating a unique digital fingerprint that ensures even the smallest modification can be detected instantly.

Blockchain-Based Integrity Protection

The generated hash is anchored on the Polygon blockchain, creating an immutable and tamper-proof record that guarantees the authenticity of digital evidence.

Secure Storage and Evidence Management

Evidence files such as images, videos, and documents are stored securely in cloud storage, while the blockchain stores the integrity proof for efficient and scalable evidence management.

Transparent Chain-of-Custody Tracking

Every transfer of evidence between authorized roles is recorded with timestamps, sender and receiver details, creating a transparent and verifiable audit trail

Intelligent Monitoring and Automated Alerts (n8n Integration)

The system integrates n8n workflow automation to detect integrity violations and trigger real-time alerts via email and WhatsApp, enabling investigators to respond immediately to potential tampering.

FLOW DIAGRAM

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Evidence Registration

- Users create cases and upload digital evidence
- System generates SHA-256 hash for each file

Blockchain Integrity

- Evidence hash stored on Polygon blockchain
- Ensures immutable and tamper-proof records

Evidence Management

- Evidence stored securely in cloud storage
- Transfers recorded for chain-of-custody tracking

Verification & Monitoring

- Evidence verified using hash comparison
- n8n automation sends tamper alerts

01 User Authentication

Secure user login with role-based access

02 Case Creation

Create investigation case and register case details

03 Evidence Upload

Upload digital evidence files into the system

04 Secure Evidence Storage

Store uploaded evidence files in cloud storage

05 Blockchain Recording

Anchor evidence hash on Polygon blockchain network

06 Hash Generation

Generate unique SHA-256 cryptographic evidence hash

07 Chain-of-Custody Tracking

Record evidence transfer between authorized roles

08 Evidence Integrity Verification

Verify file integrity using hash comparison

09 Automated Tamper Alerts

Trigger WhatsApp and email alerts via n8n

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TECHNICAL REQUIREMENTS

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Frontend Layer

Flutter Dart

- Flutter app for evidence upload, verification, QR scan, and timeline view.
- Provides role-based access for authorized users.

Backend Processing Layer

node ex

- Node.js API handling authentication, role management, and evidence processing.
- Generates SHA-256 hashes and communicates with blockchain services.

Blockchain Integrity Layer

polygon Hardhat ethers.js

- Stores evidence hashes on the Polygon blockchain
- Ensures tamper-proof and immutable evidence

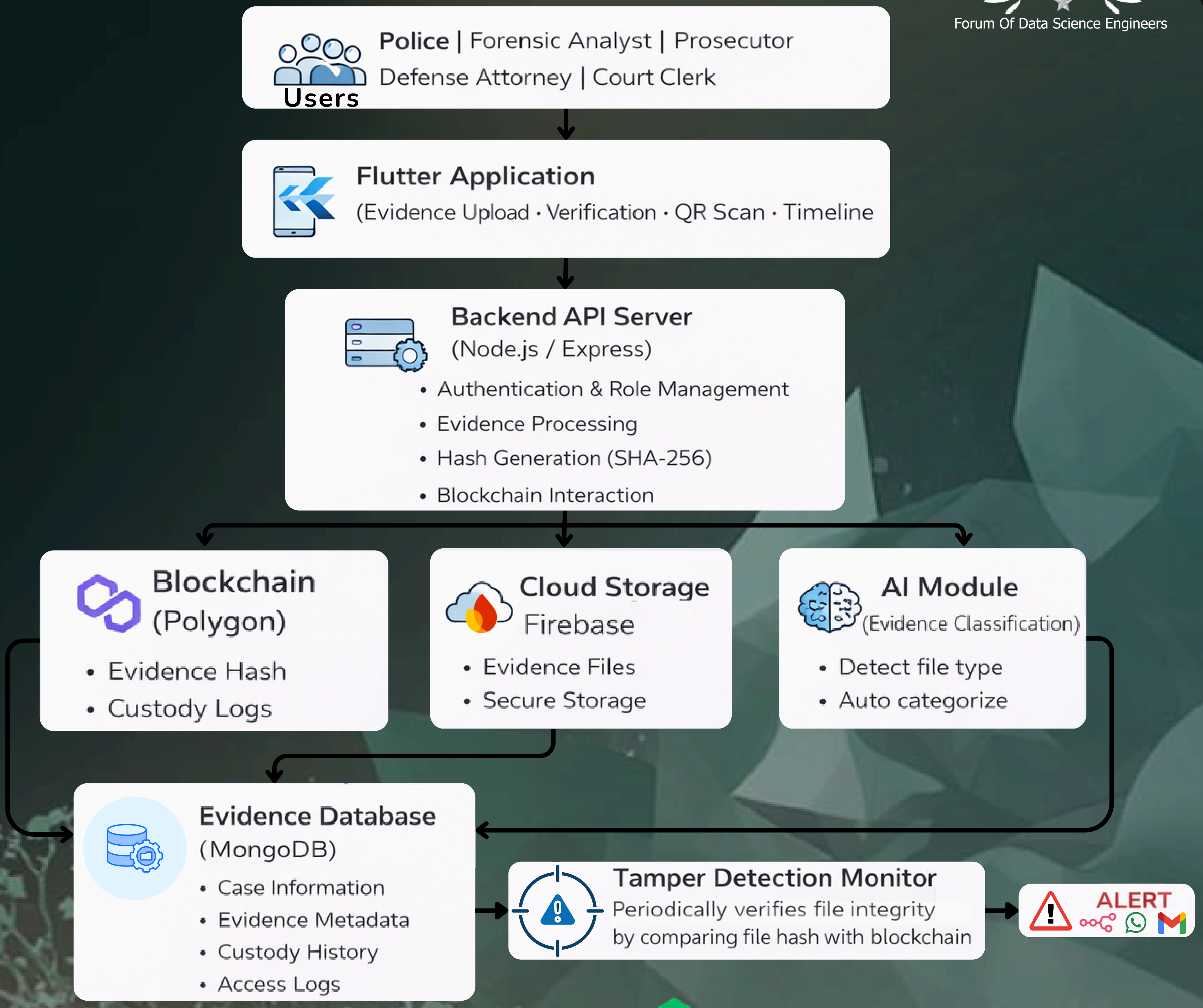
Storage, Database & AI Layer

- Cloud storage keeps evidence files securely while database stores case metadata and logs.
- AI module automatically classifies evidence (video, image, document).

Automation & Monitoring (n8n)

n8n

- n8n workflows automate system alerts and monitoring
- Sends WhatsApp and email notifications when tampering or critical events are detected



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NOVELTY / USE CASE

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Hash Integrity

01

Blockchain stores cryptographic evidence hashes ensuring immutable and tamper-proof verification.

Custody Tracking

02

Evidence transfers recorded with timestamps creating transparent and verifiable chain-of-custody.

Role Simulation

03

System simulates investigators, analysts, prosecutors, defense, and clerks managing evidence.

Instant Verification

04

Users re-upload evidence enabling automated hash comparison detecting tampering immediately.

Audit Transparency

05

Chronological logs record uploads, transfers, access attempts, and verification events.

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BUSINESS MODEL / TARGET MARKET

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Key Partnerships

- Law enforcement agencies
- Forensic laboratories
- Judicial departments
- Cloud service providers
- Blockchain infrastructure providers
- Cybersecurity organizations for security audits and threat monitoring.

Key Activities

- Digital evidence management
- Hash generation and verification
- Chain-of-custody tracking.

Key Resources

- Blockchain network (Polygon Testnet)
- Backend server and database
- Smart contracts
- Development team

Value Propositions

- Provides tamper-proof digital evidence management using blockchain.
- Ensures secure storage and verification of investigation files.
- Maintains transparent chain-of-custody tracking.
- Enables quick verification using QR codes.

Customer Relationships

- Secure role-based access system
- Dedicated support for institutions
- System training for investigators

Channels

- Government technology procurement
- Legal and cybersecurity conferences
- Institutional software deployment

Customer Segments

- Police departments
- Investigation agencies
- Forensic laboratories
- Judicial courts
- Cybercrime units
- Private forensic investigation firms that analyze digital evidence.
- Government investigation agencies

Cost Structure

- Cloud storage and hosting
- Blockchain transaction fees
- System maintenance and updates
- Security monitoring
- Development and operational costs

Revenue Streams

- Government licensing
- Enterprise subscription model
- SaaS platform access
- System integration services

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DISCOVER . DEFINE . DELIVER

TEAM DETAILS

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ROLE	NAME	STREAM/DEPT	YEAR	COLLEGE
Project Lead	SOWMIYA R	CSE	2nd	SVCE
Frontend Developer	SRILEGA R N	CSE	2nd	SVCE
Backend Developer	THIRUSHAN S R	CSE	2nd	SVCE
Blockchain Developer	VAISHNAVI CHITRAA M	CSE	2nd	SVCE
AI & Automation Developer	VEERABATHIRAN N	CSE	2nd	SVCE
Testing & Presentation	VISHAL S C	CSE	2nd	SVCE

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